

APPROVED



MSci Computing Science
School of Computing, Engineering and Technology

Version	3.0
Course Type	Undergraduate
Award Type	Master of Science - MSci
Awards	
No Course Award Assigned	
Code	0593
SCQF Level	Level 11
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	600
ECTS Credit Points	300
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100366	computer science	computer science	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	Computing Science covers an exciting blend of hardware and software topics that will enable the student to develop sophisticated applications using scientific and engineering principles. The course will provide students with a range of technical skills and knowledge needed to meet the challenges and requirements of the rapidly evolving computing industry. The student will deepen their knowledge of programming concepts and approaches, developing object-oriented software design and development skills and learn to work effectively in a variety of environments and product development technologies. In addition to providing the student with the skills that reflect the professional standards demanded by industry, the course offers a variety of specialised topics such as data science, machine learning, artificial intelligence, embedded systems, web application development, and computer security. Expertise in computing technologies and a broad range of knowledge,
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What the Course Involves	skills and personal qualities make graduates highly sought after by industry and commerce. Industrial experience can be gained on the year-long placement, or if the student chooses to Study Abroad, the experience of international study is possible. The placement is an important learning tool. The interaction with the staff of the company, the required tasks, and the general environment provides an experience difficult to achieve by other means. The student is supported on the placement by a Course Tutor from the staff of the School and an Industrial Tutor from the company. The School Tutor supports and monitors the assessment process. The course involves a broad range of teaching techniques, including lectures, seminars, tutorials, computer laboratories, industrial placement and a substantial project in the final year. The format is selected based on the suitability to the particular subject. It may consist of the staff supporting the student engage in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, etc. Generally, teaching is used to discuss the underlying theories, support the peer-learning, clarify the material and experiment with the techniques and skills discussed. This may include the use of online platforms to bring interactivity and immediacy to the teaching and learning process, using platforms such as MentiMeter for quizzes, polls, word clouds, boards, and Padlet, or Office 365 tools for collaboratively creating boards or presentations. The ability to work as an effective member of a project team is an essential skill for anyone working in industry, and group working is a feature of the course in Stages 1-3. The nature of the computing elements of all courses requires the student to gain practical experience in the use of a PC or workstation. The School of Computing (SoC) has nine laboratories housing a total of 293 PCs and 30 Macs, equipped with the latest versions of open source and commercial software for the exclusive use of SoC students. The computing lab is the venue where students, supported by staff, practise skills in the use of sophisticated software applications, and systems analysis and design tools in the laboratories. In each semester, students are expected to attend three to four hours per week per 15 credit module in lectures/tutorials/laboratories, with slightly more time being allocated in the early stages of the course. Delivery follows the University Academic Calendar , with timetabled sessions during the teaching weeks. Students are expected to continue their learning through independent study activities outside timetabled sessions. Each semester includes a Connect and Reflect Week, which is not used to deliver new academic content is delivered, but to provide opportunities for students to consolidate learning, reflect on progress and development goals, and have a concentrated focus on learning and development priorities. This course offers the opportunity for students to tailor their experience through complimentary themed pathways by elective choice, introduced in the common foundation year at Stage 1, and expanded upon from Stages 2 and 3. During Stages 2 and 3, the student will study two core Computing modules, and four subject modules that are core to the Computing Science course. The student will elect to study a subject core module from a different course within the School of Computing, to gain an appreciation for the wider context within which Computing operates and diversify their skill set. The MSci Capstone project is an extended piece of work, completed under the supervision of a staff tutor. The student agrees on a specification for a task, which may be the analysis and design of a system, the development of software or research artifact, and produces a finished thesis.
UCAS Code	Code G402
Course Collaborations	
No collaborations listed	
	Our integrated masters in Computing Science has been specifically designed to create modern software developers who can draw on their knowledge and skills to create solutions to real-world problems. This is a 5 year course that results in a Masters level qualification with a key feature of the course being is its year-long industrial, research or Study Abroad placement at Stage 4. The placement provides an unrivalled opportunity to either experience studying in another country, learning the research skills required for work in academia or

Introduction to the Course	developing knowledge and skills in a real work environment. The course is accredited by BCS, the Chartered Institute for IT for the purposes of fully meeting the academic requirement for registration as a Chartered IT Professional. The high level of expertise this degree provides, combined with the broad range of knowledge, skills and personal qualities which the School's computer science graduates possess, are highly sought after. The student will have the opportunity to learn to develop software across a number of modern platforms, including desktop, mobile, web and embedded systems and gain skills in a variety of modern programming languages. Students will be able to adapt to the changes and challenges that working with cutting edge technology brings, as well as design and test secure software systems, all in our custom built teaching labs. The computing industry needs confident, creative graduates with a wide, adaptable skillset, and that's what this course provides. The course will also provide the student with a range of technical skills and knowledge needed to meet the challenges and requirements of the rapidly changing computing industry. Providing a core focus in object-oriented software design the advanced development techniques that you learn here will form the basis of your career as a professional software developer or computer scientist. In addition to providing skills that reflect the professional standards demanded by industry, the course also offers the opportunity for the student to select to study from several specialised topics directly related to the active research and teaching areas of the School of Computing, including data science, artificial intelligence, cyber security, web and mobile design, creative design, and game design.
	Overall Aim of the Course Graduates of the MSci Computing Science course will have: • An understanding of the core concepts of the design and exploitation of computation and computer technology, including the importance of creative development strategies and innovative use of novel technologies. • The ability to apply a disciplined scientific and software engineering approach to the solution of practical problems in computing. • The ability to critically evaluate appropriate target software platforms, including an appreciation of the uncertainty, ambiguity, and trade-offs inherent in the deployment of code on these platforms, including multi-platform development approaches. • An understanding of the relevance of theory and analysis in computing, including the ability to develop and use theoretical models to abstract and simplify complex processes. • The ability to recognise the legal, social, ethical and professional issues involved in computing science and be guided by the adoption of appropriate industry standards and professional practices. • To demonstrate in-depth knowledge and understanding of a wide range of advanced and "cutting edge" topics in computing science and the ability to produce a high standard of academic documentation in specific areas of computing science research. The skills base of this course generally resides within the body of knowledge defined by the QAA Computing benchmark statement and the ACM Computer Science curriculum. As well as the technical focus, the course contains content that prepares students in social, ethical, legal and professional aspects of a cooperative human environment such as the workplace. Aims of Each Stage of the Course Stage 1 (Certificate of Higher Education) The student will undertake a common Foundation Year, to allow them to understand the breadth of Computing Science as a multidisciplinary subject, with particular focus within their chosen course. The student will acquire a knowledge and understanding of the fundamental technical concepts of programming, analysis and design of computing technologies, databases, and software development practices using appropriate tools and techniques. The student will also acquire skills, knowledge and understanding that promote continued personal and professional development through investigative, modelling, and problem solving activities that use a combination of written and computational work, both on an individual and group basis. The student will have a chance to choose to study two core modules from a different course within the School of Computing, to get an appreciation for the wider context within which Computing operates.

Educational Aims of the Course	Stage 2 (Diploma of Higher Education) The student will further develop their understanding and skills in core aspects of Computing Science. This will include advanced software design and development processes, data structures, commonly used algorithms, and dynamic full-stack web applications. The student will also develop their understanding of research, and professional skills related to entrepreneurship and collaboration. A feature of this stage is the opportunity for students to demonstrate their ability to adapt technology to address tangible, real-world problems while considering legal, ethical, and social implications. The student will also select to study two modules from different courses within the School of Computing, thereby enabling them to establish a well-rounded skill set and knowledge base that includes topics from cybersecurity, computing and creative design, or games development, as suited for their preferred career path. Stage 3 (BSc Computer Science) The student will undertake development of computer based systems for a variety of problem domains, that require the application and synthesis of advanced programming techniques and design methods. This includes learning how to use industry standard tools and frameworks in the planning, design, implementation, testing and evaluation of complex solutions appropriate for different devices. The student will learn about artificial intelligence techniques in the areas of machine learning and search. The student will further develop their innovation abilities, and approaches to cooperating with colleagues from other courses in the School of Computing, while assuming an active role in the planning and implementation of software projects, both as an individual and as an effective member of a small team. The student will again select to study two modules from different courses within the School of Computing, further developing their skills and knowledge based on their personal preferences and industry demand. Stage 4 (Placement - Study Abroad Year) The student will undertake a year-long placement working in an industry or research environment, or studying abroad at a partner institution. The student will build on their existing capabilities, achieving an in-depth knowledge and understanding of how computing and computing science techniques are used in industry or research. The student will also hone the ability to critically reflect on their own skills and competencies, a key contributor to professional development. The student will gain experience with identifying the strengths and weaknesses of approaches to problem-solving in a real-world environment, and their understanding of the wider field of computing science and its impact on society. Stage 5 (MSci Computing Science) The student will advance their knowledge of, and ability to, adapt state-of-the-art technologies in the field of artificial intelligence to craft solutions to complex problems. The student will also learn how programming languages are created and programs compiled. Throughout the second semester, the student will complete a capstone project, during which they will utilise an appropriate selection of tools, technologies, and techniques to plan, design, implement and evaluate a novel solution or research artifact for a substantial and unstructured problem with due regard to design, user experience, and professional standards.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time - 5 years, including one year placement or Study Abroad at Stage 4.
	The MSci Computing Science course is an integrated Masters enabling students to study a 5 year course and receive a Masters level degree. Uniquely Stage 4 offers the students the ability to choose between an industrial, research or Study Abroad placement. This experience allows the student to focus on their future career either by gaining extra skills and

Distinctive Features of the Course	experience in another country, in academic research skills or through industrial experience in a real workplace. The taught portion of the course has a strong practical focus on the design, implementation and testing of secure information systems, and the computer technologies that support them. A central theme of the course is multiplatform software development using multiple skills and programming languages. This approach encourages a creative development culture within the course, enabling the graduates to approach problems with confidence in the skills they have gained. The course takes a practical approach to developing the student's readiness for modern life and future employment, supporting the student with expanding the skills essential for responding to future technological and societal changes. Through the utilization of individual and group work and project teams, students are guided in cultivating skills that enhance the students' employability, such as: self-management abilities, such as focus, integrity, adaptability, and initiative; social intelligence by developing skills in communication, collaboration, and leadership; and innovation by fostering qualities such as curiosity, creativity, sense-making, and critical thinking. The course is taught by a team of full-time staff who have extensive industrial, commercial and research experience in software development and information systems. The course enables the students to benefit from their experience and involvement in the leading edge of the development and application of computer technologies to business problems. This course provides the student with the chance to customize their learning experience by selecting elective pathways focused on specific themes. This allows students to understand computing as a multidisciplinary subject and build their community of practice across the School. This option is introduced in the foundation year at Stage 1 and further developed from Stages 2 to 4. In Stage one, the student will study four core Computing modules, two subject modules core to Computing Science, and elect to study two subject core modules from different courses within the School of Computing. In Stages 2 and 3, students will engage in two core Computing modules and four subject modules integral to the Computing Science course, opting to study two subject core module from a different course within the School of Computing. In Stage 5, the student will study four subject core modules for Computing Science, culminating in the unique MSci Capstone Project in Semester 2, where the student will be supported by academics, to undertake a substantial and impactful computing-focused project that harnesses the competencies throughout their degree. The course is also accredited by the BCS allowing graduates to gain CITP status on graduation.
	MSci Industrial Placement The MSci Industrial Placement module represents work-based learning at an appropriate placement partner company. Students on this module will be able to undertake a twelve month period of paid work placement in a professional business/industrial environment. The period of supervised work placement, which precedes the final year of the MSci course will provide an important initial experience of professional practice and enhance the undertaking of further study by providing a realistic contextual awareness. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled online or in-person meetings with the student and their Industrial Supervisor during the placement. The placement period will normally run from the end of Semester 2 at Stage 3 (May/June) to the end of Semester 2 Stage 4 (May/June). This will be for a continuous period but will be assessed in two parts, one from May until December (i.e. Semester 1 Stage 4) and the other from January to May (i.e. Semester 2 Stage 4). There may be some flexibility in these times due to the range of employers and placement opportunities. To ensure the consistency of the placements, each placement will be assessed based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor, in Semester 1 and Semester 2 of Stage 4. The University provides support to students undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own

Placement/Professional Experience

connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required.

MSci Study Abroad

The MSci Study Abroad module represents learning gained whilst on an exchange with partner institutions. Students on this module will be able to undertake the equivalent of 2 Semesters of study at one of the School's partner institutions offering modules at the equivalent of SCQF level 10. Currently, this includes universities in Finland, Belgium, Malta and The Netherlands. The period of international study, which precedes the final year of the MSci course, will not only allow the students to experience learning in another country but will also provide the opportunity to develop knowledge and learn technical skills not currently taught at RGU.

Students who select to study the MSci Study Abroad module would liaise with the School's International Exchanges Coordinator to select 30 ECTS per Semester abroad at an appropriate level, equivalent to SCQF 10. This would be recorded in the student's Learning Agreement under the study abroad scheme, which is signed by the School and the host university. Based on the School's current student mobility agreements, this module can take place with at least four of the partner universities. It is expected that as current agreements are renegotiated, that appropriate modules be selected from the other partner universities.

During the period of the study abroad programme the student will be responsible to the host institution. It is expected that staff at the host institute will provide academic support for the student on the modules they are studying; however, the student will still have a Personal Tutor at the School of Computing who can provide support on other matters as needed.

The study abroad programme period will normally run from the start of Semester 1 Stage 4 (September) to the end of Semester 2 Stage 4 (May/June), although this may vary due to the differing term times in the host institutions.

At the end of the Study Abroad module the student would be required to create a portfolio of evidence, which will include their experience and transcript for their study abroad at the host institution. This portfolio will detail the knowledge and skills they have learnt and will be used to evidence the learning objectives of the MSci Study Abroad module.

MSci Research Placement

The MSci Research Placement provides the student with the opportunity to undertake a research project at the School. Students on this module will be able to undertake the equivalent of 2 Semesters of study within the School working on an active, funded, research project. The period of research work, which precedes the final year of the MSci course, will allow the student to experience the environment of working in active research. It is likely to be attractive to students who already know that they would like to pursue a career in academia. The School has previously had successful research and innovation voucher based projects that have employed students as research assistants, this module would build on this process.

Students who select to study the MSci Research Placement module would liaise with the School's Associate Dean for Research and the School's Research teams to identify current research projects with an appropriate focus for this module. Similar to the Industrial Placement, as this is an employed position, the student would be required to sit an interview to identify their suitability for the research to be undertaken.

During the period of the Research Placement the student will be responsible to the Lead Academic on the project. The research placement project period will normally run from the start of Semester 1 Stage 4 (September) to the end of Semester 2 Stage 4 (May/June), although this may vary due to the variety of projects that may be available. To ensure the consistency of the research placement each placement will be assessed in the same way as the industrial placement, i.e. through an appraisal document created by the Lead Academic and two placement reports written by the student. At the end of each Semester of the research placement, the student will submit a written report reflecting on their development while on the research project.

	Evidencing Employability In addition to studying one of the modules above, Stage 4 students will also study the 30 credit Evidencing Employability module. This module is designed to enable the student to fully reflect on and demonstrate the knowledge gained while on placement (research or industrial) or Study Abroad. More importantly, this module encourages the students to identify where this knowledge will benefit them in their future career and to provide the student with an opportunity to evidence their developing employability skills. During their placement the students will be required to keep a portfolio which will demonstrate how they have applied previous knowledge to their placement, or study abroad year. The portfolio will enable the student to identify gaps in their own skills and knowledge and to critically evaluate their own strength and weaknesses. Setting their own objectives and subsequently demonstrating the ability to either meet or plan to meet these objectives is an important aspect of this module. The module will be assessed through an evaluation of the students portfolio and a viva voce-like interview about the Stage 4 experience.
Opportunities for Further Study	As this is an integrated Masters course the award of a Masters level degree should allow entry to many doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	QAA (Quality Assessment Agency for Higher Education) published Computing Benchmarks outlining the standards expected for the award of Honours Degrees in Computing. QAA Benchmark Statements are used during the design of modules, courses and the Undergraduate Computing Programme. The structure and content of this course has been constructed following the QAA's Computing benchmarks and the QAA's principles of course design (both the current benchmarks and those described in the 2015 Benchmark Statements were consulted). The content of the course has been selected to provide a balanced coverage of the three broad categories of abilities and skills identified in the QAA Computing Benchmarks: computing-related cognitive abilities, computing-related practical abilities and additional transferable skills. These have been formulated as learning outcomes within the current Scottish Credit and Qualifications Framework (SCQF). The course design has also been strongly influenced by three key computing concepts: computational thinking, the process of analysing and developing computer systems, and the balance between theory and practice. SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide students with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's degree programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate programme. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Undergraduate Programme Management Team, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of

	computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Stage 1 Entry SQA Higher: BBBB and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. SQA Advanced Higher Year 2 Entry: BB to include either Computer Science or Maths. English at National 5 grade C or above if not held at Higher. GCE A Level Year 1 entry: BCC to include either Computer Science or Maths plus an additional numerate subject. English at GCSE grade 5/C is required if not held at A Level. Stage 2 Entry GCE A Level Year 2 entry: ABB to include Computer Science and Math. English at GCSE grade 5/C is required if not held at A Level. Irish Leaving Certificate Higher: H2H2H3H3 and two further passes to include English and Maths at grade O3 or above if not held at Higher. IB Diploma Year 1 entry: 28 to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level. IB Diploma Year 2 entry: 32 to include Math and another numerate subject at higher level pass. Standard Level English at grade 4 or above if not held at Higher Level.
Alternative Qualifications for Entry	Equivalent qualifications, e.g SQA, BTEC, S/NVQs, overseas qualifications, and access programmes will be considered on an individual basis. Applications from mature students with non-standard qualifications or work experience are welcomed.
Interview Procedures	Prospective applicants for the course are not normally interviewed.

Course-Specific Regulations	The MSci does result in the student gaining a Masters level award and as such it will be necessary to ensure a high standard of student achievement on the course. To continue on the MSci, it will be necessary for students to maintain the equivalent of a 2.1 grade average at Stage 3. Students who do not achieve this level will be permitted to transfer into the BSc Computer Science degree and complete the BSc Honours year. Students who do this will not be prevented from taking a placement year, but this placement will not be accredited towards their degree. At each stage of the MSci, student performance will be evaluated against this criteria by the Course Management Team. Students identified at Stages 1 and 2 who are in danger of not meeting the progression criteria will be counselled in the consequences of this and a support plan identified to help them progress. Students who decide not to complete the MSci placement at Stage 4 may also transfer into the BSc Computer Science degree. Likewise, students who are on the BSc (Hons) Computer Science degree who meet the progression criteria and are interested in taking the MSci placement or Study Abroad at Stage 4 may be eligible to transfer onto the MSci route at the discretion of the Course Management Team. The following modules are deemed central to the course and therefore cannot be condoned: CM4305 MSci Industrial Placement A CM4307 MSci Industrial Placement B CM4307 MSci Study Abroad A CM4308 MSci Study Abroad B CM4309 MSci Research Placement A CM4310 MSci Research Placement B CM4304 Evidencing Employability CEM321 Capstone Project CMM303 MSci Capstone Project cannot be condoned [refer Academic Regulation A4, paragraph 11]. CE2020 Short Undergraduate Placement is for additional credit only. Electives will run at the discretion of the School. Students cannot choose an elective module that they have already taken on the course, for example, some electives are aimed at particular cohorts such as articulating students. Some electives will not be available until the module is first delivered by the School, therefore, please consult the Course Leader for further information on elective availability.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.

University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	Philosophy of Course Approach to Learning The MSci Computing Science has been specifically designed to create modern software developers who can draw on their knowledge and skills to create solutions to real-world problems. Within the course there is an adherence to three principles: • To encourage innovation through investigation and understanding of computation and computing technologies. • To enhance the process of software design and development, across multiple platforms and programming languages. • To demonstrate the practical application of skills and knowledge in computing and software engineering to encourage and support creative development and problem-solving. A unique feature of this course is the Stage 4 industrial, research or Study Abroad placement. This year gives the student the opportunity to develop skills related to academic research, learn advanced skills while studying in a specialist university in Europe or gain invaluable work experience at an industrial partner.

Course-Specific Learning Experience

The course modules specifically address these ideals through the following main themes in computing science.

Multi-Platform Programming

Programming is a core skill in computer science. Rather than teach a single language on a single target platform the course provide the students with practical experience of software development across web and mobile platforms using a number of languages. This ensures that when entering employment the graduate has had exposure to a range of development experiences and development environments. This approach encourages the students to "learn to learn" enabling them to be confident in learning new languages or technologies as they develop and as required by industry.

Artificial Intelligence

Advances in Artificial Intelligence (specifically generative AI) are already impacting the practices of modern software developers. Throughout the course, the student is introduced to generative AI at relevant points in each module, with the purpose being for the student to learn how to use the technology responsibly to enhance their learning and productivity contextualised to the subject being studied. In later years of the course, the student will learn how the techniques used by AI technologies work, and to learn to develop their own AI systems. This allows the student to fully understand the capabilities and limitations of the technologies, allowing for more informed decision-making about how to utilise them effectively.

Design

This theme looks at all the aspects of design that go into building software in the modern computing industry. Including aspects of software engineering, for example, the design of requirements, code implementation, and system architectures through to the design of the user-facing components of the software or system being developed. Students will be introduced to industry standard design and prototyping tools that will support and inform these processes.

Security

An understanding of information security and the principles of secure development are a must for any modern software developer. A key concept of software development in the current environment of security threats is the identification of vulnerabilities in existing software and web systems. As part of the course, students will be shown the tools and techniques that are used to detect and secure vulnerable systems and websites. In addition, as software developers, the students will be required to understand the importance inclusion of secure code in the applications they create and will be introduced to the concept of "building in" security and defensive coding into their own work.

Professional Skills

The final theme covers the non-technical aspects of the course. Professional skills, such as project management, communication, presentation, and writing skills are extremely desirable by the computing industry. A number of modules over the course provide the students with skills in these areas and opportunities to demonstrate them, so they are fully prepared to enter the workforce on graduation. The course culminates with the Masters Capstone Project, where each student must deliver a written thesis and present a live demonstration of their work. This allows the students' to show a full complex piece of work they have designed, documented and implemented and that they can be proud of producing.

External Competitions and Challenges

As part of the course, the students are encouraged to take part in various competitions and challenges throughout the year. These challenges include:

- The School's "Hackathon" (RGUHack), where students work in teams for around 24 hours on challenges set by industry partners.
- The Google Hash Code challenge, where the students work on large optimisation challenges set by Google over a period of four hours.
- The School has also sent teams of Computer Science students to compete in the UK Cyber Security Challenges.

The Course Team strongly believe, and student feedback has shown, that competing in these events raises students' confidence in their own ability and allows them to use the knowledge and skills they have learned in a pressure, but consequence-free, environment.

	Industry Visits The course prides itself on its close links to industry, as part of this we arrange a number of opportunities for industry partners to both visit the School and host visits on their premises for our students. Technical Facilities Many of the practical aspects of the course take place in the computer laboratory. The School of Computing has 9 laboratories housing a total of 335 PCs and 24 Macs, equipped with the latest versions of open source and commercial software for the exclusive use of computing students. The School of Computing also has a range of additional hardware for use in teaching, including iPads, Raspberry Pis, photography equipment, and virtual reality headsets. Throughout the School, there is 100% wireless coverage that allows access to the University network for any student who owns a wireless-enabled laptop. The School's labs are regularly updated on a 3 year replacement cycle to ensure that the School's computing facilities are capable of delivering high performance and can run the latest software. Access to the labs is by electronic key fob at any time during University opening hours. Software provision is comprehensive, including compilers for most modern programming languages, database systems including Oracle, SQL Server and MS Office packages. On the School's PCs, standard installed software includes (amongst others) Microsoft Office 365, Adobe Creative and Production suites, Java, VS Code, Git, NodeJS, and Android development tools. Much of the software used on the course is open source or covered under an education license, allowing students to continue their work at home using the same set up as available in the labs.
	The Learning Process: Teaching Strategy and Methods The course curriculum is designed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment and supplies methodologies for the adoption of a disciplined approach to tasks such as problem definition, specification, design, implementation, and maintenance. Such knowledge includes an understanding of the range of possible options for these tasks, suitable technologies, the related practical and transferable skills including relevant approaches to collaborative work, and access to the appropriate software tools and resources. It also provides the necessary underpinning to guide practice, to ensure the sustainability of their knowledge and to provide an appropriate framework that will accommodate rapid technological change. These skills are developed at the Degree, and Masters level corresponding to the Scottish Credit and Qualifications Framework levels 7 to 11. A general feature of the course is that it stresses a unified, integrated view of the subject area. This is reflected in the fact that it is designed to: • Present a coherent underpinning of the theory appropriate to the aims of the course which is further developed and used throughout the period of study. This enables graduating students to adapt to future developments in the field. • Reflect the rapid rate of change in the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject together with their potential impact. • Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinarity within different areas of the syllabus. • Integrate modes of assessment to attempt to balance the types of assessment (individual and group) and purposes of assessment (e.g. formative and summative). • Facilitate students to exhibit competence and achievement across a number of different themes (integrated assessment). • Encourage students to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment). The professionally orientated nature of the course requires the student to obtain general and specific skills, this being addressed by a combination of teaching and learning methods,

Teaching and Learning Strategy	including lectures, tutorials, computing laboratories, workshops, seminars, practical work, optional placement, integrative project work, student-centered learning, and private study. Teaching and Learning Approaches The main goal of the teaching and learning strategy is to enable the student to become critical, reflective, independent learners who are active participants in the learning process and who develop the motivation and autonomy needed to take control and regulate their own educational experiences. This means that, throughout their course of study, the student will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their course, they will be in a position to evaluate and assess their own effectiveness. This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide range of appropriate delivery styles. These include conventional instructional teaching methodologies, social learning activities such as group- and team-working, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning. As well as the day-to-day use of the latest computer hardware and software afforded by the degree, extensive use of technology-assisted learning is made within the course of study. Examples of these include assessment by means of computer-assisted assessment packages embedded in the virtual learning environment to engage the student and increase the speed of feedback and the use of an integrated approach whereby students have short periods of instruction followed by immediate lab exercises to contextualise the topic. Modes of Delivery Lecture This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques where appropriate and the discussion of supporting ideas and consequences. The lecture is supported by a network of dissemination equipment including computer projection facilities, whiteboard, and video where appropriate. Although this type of presentation is suitable for a one-sided discourse ample opportunity exists for questions, interaction and discussion and this is facilitated by technological means. Tutorials Within the School of Computing, this format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally, this type of teaching is used to support the lecture, clarify the material and experiment with the techniques and skills required. Laboratory The nature of the computing elements of all courses requires students to gain practical skills in the use of a computer PC or workstation. This activity takes place in one of the School's computer laboratories and consists of the student, supported by staff members, practising skills in the use of sophisticated software applications and including software development and systems analysis and design tools. There has been increasing emphasis on using a "language-lab" approach to subjects in which a series of short mini-lectures are delivered in labs to allow students to gain theoretical knowledge of a topic. These are interspersed with selected practical exercises, which allow this knowledge to be immediately assimilated and so reinforce the learning process. Industrial Experience The course includes a placement, which is a useful supplement to the other teaching and learning activities of the course. The student participating in the Industrial Placement have to competitively apply for placements by submitting letters of application and CVs and by attending interviews. The student on placement undertake a variety of roles and tasks for the company they are working for, supported on the placement by a Course Tutor from the staff of the School and an Industrial Tutor from the company. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. The Academic Tutor supports and monitors the assessment
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process. In addition to the Industrial Placement, there is a series of presentations and seminars given by academic staff, staff from the careers advisory service and professionals working in key positions within the Computing and Information Technology sectors of business and industry.

Group Projects

Group projects are extended pieces of work which are usually completed under the supervision of a staff member and/or mentors within a team based module setting. Students are assigned to small project teams and given a specification for a task which may comprise elements of analysis, design and the implementation of an information system, development of a piece of software, or the exploration and practical investigation into other related activities. The students manage the project processes themselves and work as an organised team to produce a set of deliverables, which usually include a design and development component incorporating an interactive software artefact. In addition, teams are usually expected to provide accompanying documentation to a professional standard. This activity approximates closely with the development of a project undertaken by a development team within industry, and through such team based modules, these skills are practised to deliver an authentic experience which aligns with a multitude of industries.

MSci Capstone Project

This is an extended piece of work, completed under the supervision of a staff tutor. The student agrees on a specification for a task, which may be the analysis and design of a system, the development of software or similar and produces a finished thesis and product. This activity replicates the development process that would occur in an industrial development or research setting.

Student's Role in Learning

The student is expected to be an active participant in the learning process. This means that there is a requirement for the student to extend their learning from formally delivered materials through wider reading, research, and self-directed enquiry. As the student develops over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their personal objectives for learning. The student is encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which may form the basis of tutorial guidance with staff. The student is expected to utilize the full range of facilities and resources available to them, for example, the services and resources provided by the library.

The Assessment of Student Work

A variety of assessment methods are used in this course, as deemed appropriate to the subject material being assessed and the context in which the material is delivered. The University has developed an Assessment and Feedback Standard, which guides all assessment practices on the course. This ensures the student has opportunities to engage with assessment and feedback which extends and enhances their knowledge and skills; offers a balanced and manageable assessment load; and provides good quality, effective feedback. Assessments are developed so that they ensure module and course-wide learning outcomes have been met corresponding at the appropriate SCQF Level 7-10 for the course stage with respect to the characteristics of knowledge and understanding; practical skills; generic cognitive skills; communication, numeracy and IT skills, autonomy, accountability and working with others.

It is important to note that all grades indicated to the student during the course of the academic session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal.

The School Office prepares a schedule of coursework assessment and feedback deadlines. The assessment timetable is structured to ensure workload and deadline tensions are suitably managed. The student is informed how their assessment will be scheduled and the criteria made transparent in the context of coursework and grading structures.

Assessment

Industry, Research, or Study Abroad Placement

The student on industry or research placement is assessed by Tutor and Supervisor assessment. Those on Study Abroad are assessed at the host institution, and by RGU Tutor assessment.

Computer Aided Assessment

Computer-assisted assessment packages integrated into the virtual learning environment such as Moodle's question system are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback.

Peer Assessment

Peer assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop working and interpersonal skills, as well as honing appraisal and critique skills. Peer assessment may be used in both summative and formative contexts.

Unseen Exams

In general unseen exams are used to assess the student's knowledge and understanding, and as the student progresses through the course unseen exams feature as a component of the assessment of taught modules.

Assessed Laboratories

Assessed laboratories are used to assess the student's knowledge, understanding and practical skills, and generally involve the student undertaking an unseen practical task such as developing a part of an information system, under open or closed book, supervised conditions with a fixed time constraint.

Coursework

For summative assessment the achievement of Learning Outcomes is assessed by coursework or a combination of exam and coursework. Coursework may take many forms, i.e. reports, essays, production of software, portfolio of work, presentations, design documentation and modelling activities.

Projects/Dissertations

Extended projects are used in later stages as a learning vehicle and assessment process that allow the student to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. These projects often require the practical application of a number of different techniques and technologies and provide the students with the opportunity to learn new investigative and evaluative skills. Projects, in particular group projects, provide an opportunity to learn new investigative and evaluative skills, and are also often used as the vehicle for gaining valuable employability skills. Project work is usually assessed by a combination of assessment of the completed project deliverable as an artefact, accompanying report or dissertation, and verbal presentation and demonstration documenting the project work undertaken by the student.

Assessment Types

Some modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity (e.g. individual and group activities) and purposes of assessment (e.g. formative and summative).

Where there are a number of modules that cover allied subject material, integrated assessment may be used to facilitate the student to exhibit competence and achievement across a number of different themes.

In all years, synoptic assessments embedded into projects and other practical assignments are used to encourage the student to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area.

	Equality and Diversity Any datasets used within the course will aim to adhere to the Software Engineering Code of Ethics and Professional Practice recommended by the ACM/IEEE-CS Joint Task Force which states that "only accurate data derived by ethical and lawful means" will be used, and that its integrity will be maintained. Furthermore, the student will be taught about recognising and researching known bias within datasets, and where possible, will be signposted to use datasets that are less likely to have inherent bias.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	In addition to the University-wide support for Teaching and Learning, the school provides a number of means of supporting the student in teaching and learning. Members of staff publish course based teaching and learning materials on the University's Virtual Learning Environment, Moodle. The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for the student, such as codes of conduct in laboratories. Information on studying abroad scheme can be accessed at http://www.rgu.ac.uk/future-students/study-abroad . Every student in the school is provided with an academic member of staff who will act as their Personal Tutor throughout their studies. The personal tutor system is there to ensure that every student has a named point of contact to seek advice on a range of issues, both personal and academic, that may be of concern to the student or affecting their studies. The school runs a comprehensive induction programme during freshers week where the student will be introduced to the school, introduced to key members of staff including their Course Leader and their Personal Tutor. During the induction event, the student will be made familiar with the University's IT, learning environment and associated systems. The students can also be provided with keyfob access to dedicated labs and other spaces in which teaching takes place. Feedback is provided continuously to the student through informal contact with subject lecturers and tutors in the tutorial and laboratory settings. Feedback is provided on assessed practical work within 20 working days of submission of the work. Feedback on examination results can be provided by module staff, and facilities are made available for all students to

	view their marked examination scripts on request and under supervision. Grades are made available electronically for viewing via the student portal following ratification by the Assessment Board. The 'Student Handbook' provides detail on the School, course, and working environment. It contains information regarding teaching and assessment, IT support, student support services, learning resources including the library and health & safety. It details key contact names and email addresses. The handbook will be located on Moodle and will be organised into sections so that information is readily accessible.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	This course fully complies with the above procedures. The course management team meets regularly to review any specific course issues, student feedback and module evaluation, External Examiners' reports and matters arising from Staff Student Liaison meetings. These meetings will include representatives from the School of Computing. Annual Course /Programme Appraisals are reviewed and approved by the Programme Management Team and School Academic Board which meet at least once per semester. Within the School of Computing, the School's Teaching Committee regularly looks at module metrics such as student experience and student achievement to ensure that the models are continuing to deliver the required standard set by the school. The School's Teaching Committee, the Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the course. The Undergraduate Computing Programme is also informed by feedback from the School's Industrial Board.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements.

University Student Feedback Mechanisms

This approach is underpinned by the *Student Partnership Agreement* which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: [Student Representation & Partnership | RGU](#) .

There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

Mechanisms through which students are engaged and supported include:

- participation in student evaluation questionnaires and where relevant the *National Student Survey (NSS)* ;
- staff/student liaison arrangements at course/programme level;
- support for Student Representatives;
- student representation on Institution-Led Subject Review Panels and Validations;
- student representation on key University committees.

Course-Specific Student Feedback Mechanisms

Within the School of Computing, there is a single Staff/Student Liaison Committee for all undergraduate degree courses. It normally consists of the Dean of School, Associate Dean for Academic Development and Student Experience and Course Leaders, with student representatives from each stage of each course. The Committee meets at least once per semester and feeds appropriate information to other departments and/or committees as required.

Student representatives also take part in the School Academic Board where they can contribute to discussions about School policy. In addition, there are regular scheduled meetings between the student and Personal Tutors, and Course Leaders and student representatives to discuss issues arising within specific classes and on individual modules.

Student Voice Questionnaires are distributed to students on all modules during semester which allow them to raise issues and give feedback to staff.

Awards**Stage 1**

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Certificate of Higher Education - CertHE	Computing	Aberdeen, United Kingdom	Exit Award		120

Stage 2

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Diploma of Higher Education - DipHE	Computing	Aberdeen, United Kingdom	Exit Award		240

Stage 3

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<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Science - BSc	Computer Science	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>
No awards exist

Stage 5

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Master of Science - MSci	Computing Science	Aberdeen, United Kingdom	Target Award		600

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate a basic knowledge and understanding of the basic context of computer science, and its relationship to other subject areas such as mathematics, science, and engineering.	SC1	SCQF 7
LLO2	Demonstrate familiarity with the basic facts and principles of computing and supporting topics via demonstrations of skills gained, the ability to answer questions and to describe and employ basic tools and techniques. The student is introduced to the basic skills and knowledge required for operating in a computer environment via study and practice.	SC1	SCQF 7
LLO3	Outline understanding, at an appropriate, elementary level of programming and the development of software, computer hardware, and show an understanding of the business environment and roles played by information systems in organisations.	SC1	SCQF 7
LLO4	Obtain a working understanding of basic facts and principles in computing and information security via reports, solutions to standard problems, and projects on fundamentals an appropriate use of conventions via the topics outlined above.	SC1	SCQF 7
LLO5	Carry out study of the subject in a systematic, directed way with the aid of appropriate tutor guidance within the classroom and laboratory.	SC2	SCQF 7
LLO6	Apply the knowledge, skills, and understanding of computer science, web development, and database systems in practical contexts.	SC2	SCQF 7
LLO7	Employ some of the basic and routine professional skills, techniques, and practices associated with computing.	SC2	SCQF 7
LLO8	Apply these skills in both routine and non-routine contexts.	SC2	SCQF 7
LLO9	Apply the basic principles and procedures found in computing science to standard, simple situations.	SC3	SCQF 7
LLO10	Present the ability to analyse situations and problems, based on the set of classifications presented by tutors.	SC3	SCQF 7
LLO11	Systematically relate to a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO12	Start to form their own value judgments based on the criteria provided.	SC3	SCQF 7
LLO13	Apply the basic tools/methods of computing science to simple, standard problems.	SC3	SCQF 7
LLO14	Convey aspects of computing science in a clear and concise way, in writing and orally, in relatively informal and limited-length pieces of work. Students should also be able to communicate appropriately the basic design and documentation of programs.	SC4	SCQF 7
LLO15	Achieve basic numeracy, mathematics and the ability to manipulate data at a level appropriate to this stage of the degree. Students should also develop an understanding of basic concepts in logic and computation.	SC4	SCQF 7
LLO16	Obtain practical skills in computing science using basic ICT applications as well as those skills necessary for efficient use of computer systems.	SC4	SCQF 7
LLO17	Manage learning in an increasingly effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities at a professional level of practice.	SC5	SCQF 7
LLO18	Carry out interactions effectively with tutors and fellow students via software and web development tasks, problem-solving tutorials and the production of appropriate documentation, and accept supervision in less familiar areas of work.	SC5	SCQF 7
LLO19	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure while managing limited resources within defined areas of work.	SC5	SCQF 7
LLO20	Take the lead in implementing agreed plans in familiar or defined contexts, while taking account of own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO21	Work, under guidance, with others to acquire an understanding of current professional practice.	SC5	SCQF 7

LLO22	Report an understanding of the important central areas of this stage of the course, specifically, software development, dynamic web development, collaborative and professional skills, and research methods.	SC1	SCQF 8
LLO23	Infer a growing appreciation of the scope of the field of study so enabling students to map knowledge into a coherent picture of the discipline.	SC1	SCQF 8
LLO24	Compare the significance of central and peripheral areas of the subject, suggesting opportunities to explore and extend the boundaries of the discipline.	SC1	SCQF 8
LLO25	Practice knowledge, skills, and understanding in using a range of professional skills, techniques, and practices associated with the computing topics studied, some of which are at an advanced or complex level	SC2	SCQF 8
LLO26	Show the ability to carry out routine lines of enquiry, development or investigation into professional level problems and issues.	SC2	SCQF 8
LLO27	Adapt routine practices within accepted standards.	SC2	SCQF 8
LLO28	Use standard principles and procedures to somewhat more demanding situations.	SC3	SCQF 8
LLO29	Demonstrate increasing ability to apply classifications and critically analyse relatively simple situations.	SC3	SCQF 8
LLO30	Demonstrate an appreciation of the need for relating and marshalling a range of facts/ideas/elements in an argued case.	SC3	SCQF 8
LLO31	Compare a range of valid alternative solutions, and begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way, particularly within the context of group work.	SC3	SCQF 8
LLO32	Use given tools/methods accurately and carefully to more demanding problems appropriate to this stage.	SC3	SCQF 8
LLO33	Use communication skills in more formal academic and professional styles, and in more extended pieces of work.	SC4	SCQF 8
LLO34	Undertake the application of a selection of more advanced numerical, modelling and design skills, relevant to specialised areas of software development, dynamic web programming, and research.	SC4	SCQF 8
LLO35	Use a broad selection of more advanced computing and IT skills in a wide range of contexts in all topics, but particularly in the development of solutions to computing problems.	SC4	SCQF 8
LLO36	Practice effective interactions within a learning- or subject-specific group, demonstrating basic negotiating, role-ownership, leadership and group-support skills.	SC5	SCQF 8
LLO37	Plan work activities in familiar or defined contexts.	SC5	SCQF 8
LLO38	Practice management of resources and exercise responsibility for the work of others within a group, within a defined structure. Practise in ways that show awareness of own and others' roles, responsibilities, and contributions when carrying out and evaluating tasks.	SC5	SCQF 8
LLO39	Work, under guidance, with others to acquire an understanding of current legal, ethical and professional issues, and manage these issues in accordance with current professional and/or ethical codes of practices.	SC5	SCQF 8
LLO40	Discuss a knowledge and understanding of the important central areas of this stage of the degree; entrepreneurship, project management, software development, and artificial intelligence.	SC1	SCQF 9
LLO41	Critically explain a range of the principles, main theories, concepts and terminology of computing science.	SC1	SCQF 9
LLO42	Confidently discuss the broad areas of the knowledge/subject bases of computing science, including key specialist areas.	SC1	SCQF 9
LLO43	Explain the limits of the broad area of study, together with an ability to relate the different areas to one another in a systematic and logical manner.	SC1	SCQF 9
	Demonstrate knowledge, skills, and understanding in using a range of the principal professional skills, techniques, practices and/or materials		

LLO44	associated with computing science.	SC2	SCQF 9
LLO45	Demonstrate an applied knowledge of some specialised or advanced skills, techniques, practices and/or materials in the implementation of computing science/information systems solutions.	SC2	SCQF 9
LLO46	Demonstrate application of routine methods of enquiry and research.	SC2	SCQF 9
LLO47	Demonstrate ability to work in a range of professional contexts that include a degree of unpredictability.	SC2	SCQF 9
LLO48	Demonstrate the ability to select and use principles and procedures appropriate to the situation or problem in hand, particularly within software project management.	SC3	SCQF 9
LLO49	Point out familiar ideas or principles in new contexts or situations; analyses systematically and effectively.	SC3	SCQF 9
LLO50	Assemble facts/ideas/elements in an argued case; produces new design ideas in a wider range of situations, particularly within software project management.	SC3	SCQF 9
LLO51	Differentiate among a range of valid alternative solutions; begins to discriminate and evaluate in a reasoned, systematic and increasingly independent way particularly within software project management.	SC3	SCQF 9
LLO52	Draw on established skills to begin to self-direct problem-solving, identify key elements or problems, and choose appropriate methods to construct their resolution, with only general guidance in all topics.	SC3	SCQF 9
LLO53	Effectively interpret communication requirements for a variety of roles, to produce clear, well-structured reports and presentations on relevant aspects of computer science.	SC4	SCQF 9
LLO54	Defend the selection of specialist numerical, modelling and design skills, relevant to the development and implementation of secure information systems, and the management and software engineering projects.	SC4	SCQF 9
LLO55	Access and successfully use a broad selection of specialist computing and ICT skills in a wide range of contexts, in all topics.	SC4	SCQF 9
LLO56	Demonstrate a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO57	Assume an appropriate role in the design and management of secure information systems development projects.	SC5	SCQF 9
LLO58	Demonstrate ability to use negotiation and collaboration skills to manage the interface between computing, information security and other functional areas of an organisation, and the users of its computer system	SC5	SCQF 9
LLO59	Assess appropriate response to legal, ethical and professional issues in accordance with current codes of best practice, seeking guidance where appropriate.	SC5	SCQF 9
LLO60	Illustrate an understanding of the context of, and the relationship between, the principal areas, features, boundaries, terminology, and conventions of computer science practice used in the industrial placement, research placement or study abroad placement.	SC1	SCQF 10
LLO61	Illustrate confident familiarity and critical understanding of the defining concepts and features of computing science as used in the industrial placement, research placement or study abroad placement.	SC1	SCQF 10
LLO62	Develop appropriate high-level expertise in relevant mathematical modelling, data analysis; and implementation skills, and apply specialist analysis within the industrial placement, research placement or study abroad placement while learning technical skills appropriate to the specific task.	SC1	SCQF 10
LLO63	Communicate detailed knowledge and understanding in one or more computing science specialisms which are appropriate to the industrial placement, research placement or study abroad placement.	SC1	SCQF 10

LLO64	Communicate a clear understanding of their subject's boundaries, of the limits of its application, in particular in an industrial or research arena.	SC1	SCQF 10
LLO65	Execute knowledge, skills, and understanding in using a wide range of the principal professional skills, techniques, and practices associated with computing science either in the industrial placement, research placement or study abroad placement.	SC2	SCQF 10
LLO66	Illustrate practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of the subject of computing science, as used in the industrial placement, research placement or study abroad placement.	SC2	SCQF 10
LLO67	Illustrate application of practical knowledge of computing science to execute a defined project while on the industrial placement, research placement or study abroad placement.	SC2	SCQF 10
LLO68	Operate in a range of professional level contexts that include a greater degree of unpredictability and specialism in computing science.	SC2	SCQF 10
LLO69	Confidently compose an accurate selection and application of principles and procedures to the solution of a range of professional situations and problems while on the industrial placement, research placement or study abroad placement.	SC3	SCQF 10
LLO70	Examine and classify principles and ideas in new contexts and situations, and analyse them systematically, effectively and critically in terms of their importance and application to the industrial placement, research placement or study abroad placement.	SC3	SCQF 10
LLO71	Compose facts, ideas, and elements in support of a well-structured argument, design novel solutions and evolve new concepts with respect to the needs of the placement host organisation.	SC3	SCQF 10
LLO72	Reconcile theory with professional and vocational practice in the industrial placement, research placement or study abroad placement, autonomously evaluate theory, process, solutions, and outcomes critically and effectively.	SC3	SCQF 10
LLO73	Develop creative solutions to complex problems related to the industrial placement, research placement or study abroad placement needs in an increasingly independent, confident and flexible way and apply knowledge and skills appropriate to their solution, at the threshold of professional competence in all topics.	SC3	SCQF 10
LLO74	Justify the use and access of a broad selection of specialist computing, software engineering, and information systems skills, appropriate to the industrial placement, research placement or study abroad placement.	SC4	SCQF 10
LLO75	Communicate their effective use of specialist computing and ICT facilities concerned with the software development and information systems used by the industrial placement, research placement or study abroad placement and develop appropriate skills for this purpose.	SC4	SCQF 10
LLO76	Engage effectively with colleagues or peers whilst on the industrial placement, research placement or study abroad placement, to express complex ideas to colleagues through appropriate media, based on considerations of context.	SC4	SCQF 10
LLO77	Engage effectively in independent roles, debate in a professional manner, and produce detailed critiques and coherent project reports appropriate to the industrial placement, research placement or study abroad placement.	SC4	SCQF 10
LLO78	Communicate confidently with informed audiences in a range of contexts via oral and formal presentations in a wide range of contexts.	SC4	SCQF 10
LLO79	Critique the demonstration of skills, knowledge, and understanding through autonomous investigative, modelling and problem-solving activities, and manage the process relating to the development, maintenance and review of secure information systems.	SC5	SCQF 10
LLO80	Argue the extent to which they maintain standards and quality control as applied to the specification, production, implementation, and maintenance of secure information systems, in accordance with current professional and ethical codes of practices.	SC5	SCQF 10
LLO81	Critique their own approach to independent learning when provided with minimal guidance using a wide range of resources appropriate to the profession, seeking and making effective use of feedback from colleagues or staff at the industrial placement, research placement or study abroad placement.	SC5	SCQF 10
	Interact effectively within groups and other staff or peers in all topics, and, as appropriate to the industrial placement, research placement or		

LLO82	study abroad placement, and develop interaction and negotiating skills with customers, clients and other stakeholders.	SC5	SCQF 10
LLO83	Appraise their understanding of the context of, and the relationship between, the principal areas, features, boundaries, terminology, and conventions of specialist areas of the course. The specialist areas at this level are; compilers and language development, artificial intelligence, and research	SC1	SCQF 11
LLO84	Analyse their familiarity and critical understanding of the defining concepts and features of their chosen area of advanced computing science areas.	SC1	SCQF 11
LLO85	Construct appropriate high-level expertise in relevant mathematical modelling, data analysis, machine learning, artificial intelligence; and implementation skills, and apply specialist analysis within the MSci Capstone Project.	SC1	SCQF 11
LLO86	Prepare demonstrations of a detailed knowledge and understanding in one or more computer science specialisms, some of which are informed by, or at the forefront of the subject.	SC1	SCQF 11
LLO87	Criticise their awareness of current issues and research areas in computing science.	SC1	SCQF 11
LLO88	Create knowledge, skills, and understanding in using a significant range of the principal professional skills, techniques, and practices associated with computing science.	SC2	SCQF 11
LLO89	Create solutions using a range of practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of the subject of computing science.	SC2	SCQF 11
LLO90	Appraise their use of a practical knowledge of computing science to execute a significant project of research and development in the MSci Capstone Project module.	SC2	SCQF 11
LLO91	Produce work in a range of professional level contexts that include a greater degree of unpredictability and specialism in computing science.	SC2	SCQF 11
LLO92	Critically analysis, evaluate, and synthesise computing science research, particularly within the MSci Capstone Project module.	SC3	SCQF 11
LLO93	Analyse and define new and abstract problems and issues and analyse them systematically, effectively and critically, particularly within the MSci Capstone Project module.	SC3	SCQF 11
LLO94	Construct well-structured arguments by bringing together supportive facts, ideas, and elements, develop original and creative responses to problems and issues, working autonomously, particularly within the MSci Capstone Project module.	SC3	SCQF 11
LLO95	Synthesise theory with professional and vocational practice, autonomously evaluate theory and extend knowledge, skills, and practices, and thinking in computing science.	SC3	SCQF 11
LLO96	Prepare definitions of complex problems in an independent, confident and flexible way and apply knowledge and skills appropriate to their solution, deal with complex issues and make informed judgements in situations in the absence of complete or consistent data/information.	SC3	SCQF 11
LLO97	Evaluate their demonstration, use and access of a wide range of specialist and advanced computing, software engineering, and information systems skills, in all topics, but particularly within the MSci Capstone Project module.	SC4	SCQF 11
LLO98	Prepare communications, using appropriate methods, to express complex ideas to a range of audiences with different levels of knowledge /expertise.	SC4	SCQF 11
LLO99	Make informed judgements on how to engage effectively in independent roles, debate in a professional manner with peers and specialists and produce detailed critiques and coherent project reports.	SC4	SCQF 11
LLO100	Confidently prepare and give confident oral and formal presentations, including the use of graphical and numerical data to a wide range of audiences in a wide range of contexts, particularly within the MSci Capstone Project module.	SC4	SCQF 11

LLO101	Take responsibility for their demonstration of skills, knowledge, and understanding by exercising substantial autonomy and initiative in investigative, modelling and problem-solving activities, and manage the process relating to the development, maintenance and review of secure information systems.	SC5	SCQF 11
LLO102	Appraise complex ethical and professional issues and make informed judgments on issues not addressed by current professional and/or ethical codes or practices.	SC5	SCQF 11
LLO103	Take responsibility for their own learning, with significant responsibility for their own resources appropriate to the problem or investigation area. At this level within all topics, but particularly within the project.	SC5	SCQF 11
LLO104	Make informed judgements on taking leadership and/or initiative and make an identifiable contribution to computing science research or development practices.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9

SC5	Autonomy, accountability and working with others	SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
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Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Stage 1 / Semester 1**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1109	Exploring Computing Devices		Tiffany Young (ty7997)	Yes	2.0	15	Level 7
Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 1	Computing Science - Electives Group 1		Laura Tucker		1.0	15	

Stage 1 / Semester 2**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7
Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 2	Computing Science - Electives Group 2		Laura Tucker		1.0	15	

Stage 2 / Semester 1**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2115	Advanced Software Design and Development		Mark Snaith (ms10541)	Yes	1.0	15	Level 8
Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 3	Computing Science - Electives Group 3		Laura Tucker		3.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2116	Data Structures and Algorithms		David Corsar (dc9396)	Yes	2.0	15	Level 8
Full								

Time On Campus	CE2001	Full Stack Applications		David Corsar (dc9396)	Yes	1.0	15	Level 8
Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 4	Computing Science - Electives Group 4		Laura Tucker		3.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8

Stage 3 / Semester 1**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3113	Concurrent Programming		David Lonie (dpl0795)	Yes	2.0	15	Level 9
Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
Full Time On Campus	CM3151	Machine Learning		David Corsar (dc9396)	Yes	1.0	15	Level 9

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 5	Computing Science - Electives Group 5		Ross McLean (rjm7342)		2.0	15	

Stage 3 / Semester 2**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3152	Artificial Intelligence for Problem Solving		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 9
Full Time On Campus	CM3142	Internet of Things		David Corsar (dc9396)	Yes	1.0	15	Level 9
Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0593 - Electives 6	Computing Science - Electives Group 6		Laura Tucker		2.0	15	

Stage 3 / Semester 3**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8

Stage 4 / Semester 1**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4304	Evidencing Employability	Undertaken over two semesters.	Mark Bartlett (mb9400)	Yes	1.0	30	Level 10
Full Time On Campus	CM4305	MSci Industrial Placement A	Route A	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
Full Time On Campus	CM4309	MSci Research Placement A	Route C	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
Full Time On Campus	CM4307	MSci Study Abroad A	Route B	Mark Zarb (mz7371)	Yes	1.0	45	Level 10

Stage 4 / Semester 2**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4306	MSci Industrial Placement B	Route A	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
Full Time On Campus	CM4310	MSci Research Placement B	Route C	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10

Full Time On Campus	CM4308	MSci Study Abroad B	Route B	Mark Zarb (mz7371)	Yes	1.0	45	Level 10
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Stage 4 / Semester 3**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8

Stage 5 / Semester 1**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CMM302	Computing Science Research		Laura Tucker	Yes	2.0	15	Level 11
Full Time On Campus	CMM304	Knowledge Modeling and Reasoning		David Corsar (dc9396)	Yes	2.0	15	Level 11
Full Time On Campus	CMM306	Languages and Compilers		Mark Bartlett (mb9400)	Yes	3.0	15	Level 11
Full Time On Campus	CEM300	Natural Language Processing		Mark Bartlett (mb9400)	Yes	1.0	15	Level 11

Stage 5 / Semester 2**Full Time On Campus**

Route A - Industrial Placement. Route B - Study Abroad. Route C - Research Placement. Please note CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	CM1109	Exploring Computing Devices		Tiffany Young (ty7997)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
1	Semester 1	Elective	Full Time On Campus	PS0593 - Electives 1	Computing Science - Electives Group 1		Laura Tucker		1.0	15	
1	Semester 2	Core	Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7

1	Semester 2	Core	Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
1	Semester 2	Elective	Full Time On Campus	PS0593 - Electives 2	Computing Science - Electives Group 2		Laura Tucker		1.0	15	
2	Semester 1	Core	Full Time On Campus	CM2115	Advanced Software Design and Development		Mark Snaith (ms10541)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
2	Semester 1	Elective	Full Time On Campus	PS0593 - Electives 3	Computing Science - Electives Group 3		Laura Tucker		3.0	15	
2	Semester 2	Core	Full Time On Campus	CM2116	Data Structures and Algorithms		David Corsar (dc9396)	Yes	2.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CE2001	Full Stack Applications		David Corsar (dc9396)	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
	Semester		Full Time								

2	2	Elective	On Campus	PS0593 - Electives 4	Computing Science - Electives Group 4		Laura Tucker		3.0	15	
2	Semester 3	Elective	Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
3	Semester 1	Core	Full Time On Campus	CM3113	Concurrent Programming		David Lonie (dpl0795)	Yes	2.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3151	Machine Learning		David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 1	Elective	Full Time On Campus	PS0593 - Electives 5	Computing Science - Electives Group 5		Ross McLean (rjm7342)		2.0	15	
3	Semester 2	Core	Full Time On Campus	CM3152	Artificial Intelligence for Problem Solving		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3142	Internet of Things		David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	PS0593 - Electives 6	Computing Science - Electives Group 6		Laura Tucker		2.0	15	

3	Semester 3	Elective	Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
4	Semester 1	Core	Full Time On Campus	CM4304	Evidencing Employability	Undertaken over two semesters.	Mark Bartlett (mb9400)	Yes	1.0	30	Level 10
4	Semester 1	Core	Full Time On Campus	CM4305	MSci Industrial Placement A	Route A	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
4	Semester 1	Core	Full Time On Campus	CM4309	MSci Research Placement A	Route C	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
4	Semester 1	Core	Full Time On Campus	CM4307	MSci Study Abroad A	Route B	Mark Zarb (mz7371)	Yes	1.0	45	Level 10
4	Semester 2	Core	Full Time On Campus	CM4306	MSci Industrial Placement B	Route A	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
4	Semester 2	Core	Full Time On Campus	CM4310	MSci Research Placement B	Route C	Mark Bartlett (mb9400)	Yes	1.0	45	Level 10
4	Semester 2	Core	Full Time On Campus	CM4308	MSci Study Abroad B	Route B	Mark Zarb (mz7371)	Yes	1.0	45	Level 10
4	Semester 3	Elective	Full Time On Campus	CE2020	Short Undergraduate Placement		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
	Semester		Full Time								Level

5	1	Core	On Campus	CMM302	Computing Science Research		Laura Tucker	Yes	2.0	15	11
5	Semester 1	Core	Full Time On Campus	CMM304	Knowledge Modeling and Reasoning		David Corsar (dc9396)	Yes	2.0	15	Level 11
5	Semester 1	Core	Full Time On Campus	CMM306	Languages and Compilers		Mark Bartlett (mb9400)	Yes	3.0	15	Level 11
5	Semester 1	Core	Full Time On Campus	CEM300	Natural Language Processing		Mark Bartlett (mb9400)	Yes	1.0	15	Level 11
5	Semester 2	Core	Full Time On Campus	CEM321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf A number of external sources were consulted in the process of curriculum development for the MSci Computing Science course. The accreditation guidelines for the BCS, The Chartered Institute for IT, Guidelines on course accreditation, Information for universities and colleges, January 2020, can be found at https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf and Academic Accreditation Guidelines, April 2022 can be found at https://www.bcs.org/media/1209/accreditation-guidelines.pdf The Association for Computing Machinery (ACM) Guidelines on curricular elements for Computer Science (and related areas), Computer Science Curricula 2020, can be found at https://www.acm.org/binaries/content/assets/education/curricula-recommendations/cc2020.pdf